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Batch-B3

Experiment no.1

```
#include<stdio.h>
#include<conio.h>

int length(char[]);
void compare(char[], char[]);
void palindrom(char[]);
void substring(char[], char[]);
void copy(char[], char[]);    // Copy second string into first
void reverse(char[]);
void displayMenu();
int i,j;

int main(){
int choice;
char str[50];
char str2[50];
int exit = 0;
displayMenu();
do{
printf("ENTER CHOICE (7 for Display Menu): ");
scanf("%d", &choice);

if(!(choice == 7 || choice == 8))
{
   printf("Enter String: ");
   scanf("%s", str);
}
switch(choice)
{
   case 1:
      printf("Length of the String is %d\n\n", length(str));
      break;
```

```
    case 2:
       printf("Enter second String: ");
       scanf("%s", str2);
       copy(str, str2);
       break;
    case 3:
       reverse(str);
       break;
    case 4:
       palindrom(str);
       break;
    case 5:
       printf("Enter substring String: ");
       scanf("%s", str2);
       substring(str, str2);
       break;
    case 6:
       printf("Enter second String: ");
       scanf("%s", str2);
       compare(str, str2);
       break;
    case 7:
       displayMenu();
       break;
    case 8:
       exit = 1;
       break;
    default:
       printf("Invalid Choice!\n");
}
}
while(exit == 0);

return 0;
}
void displayMenu()
{
    printf("<<----------Menu---------->>\n");
    printf("1. Length\n");
    printf("2. Copy String into another String\n");
```

```
    printf("3. Reverse String\n");
    printf("4. Check for Palindrome\n");
    printf("5. Check for Substring\n");
    printf("6. Compare Any String with Another\n");
    printf("7. Display Menu\n");
    printf("8. Exit\n");
}
//Calculate length of the string
int length(char a[])
{
    int len = 0;
    while(a[len] != '\0')
    {
        len++;
    }
    return len;
}
//Compare the strings
void compare(char str[50], char str2[50])
{
    i = 0;

    while(str[i] != '\0' && str2[i] != '\0')
    {
        if(str[i] != str2[i])
        {
            printf("Both Strings are Not Equal\n\n");
            return;
        }
        i++;
    }

    if(str[i] == '\0' && str2[i] == '\0')
    {
        printf("Both Strings are Equal\n\n");
    }
    else
    {
        printf("Both Strings are Not Equal\n\n");
    }
```

```
}
//Palindrome function
void palindrom(char a[])
{
   int l = length(a);
   int flag = 0;

   for(i=0, j=l-1; i<j; i++, j--)
   {
      if(a[i] != a[j])
      {
         flag = 1;
         break;
      }
   }

   if(flag == 0)
      printf("String is Palindrome\n\n");
   else
      printf("String is Not Palindrome\n\n");
}
//Check for substring
void substring(char a[50], char b[50])
{
   int al = length(a);
   int bl = length(b);
   int flag = 0;
   int count = 0;
   if(bl <= al)
   {
      for(i=0; i<=al-bl; i++)
      {
         count = 0;

         for(j=0; j<bl; j++)
         {
            if(a[i+j] == b[j])
            {
               count++;
            }
```

```
            else
            {
               break;
            }
         }
         if(count == bl)
         {
            flag++;
         }
      }
      if(flag == 0)
      {
         printf("The String B is not substring of A\n\n");
      }
      else
      {
         printf("The String B is substring of A and it occur %d times\n\n", flag);
      }
   }
   else
   {
      printf("The String B is not substring of A\n\n");
   }
}
//Copy string
void copy(char a[], char b[])
{
   int al = length(a);
   int bl = length(b);
   char str3[50];

   for(i=0; i<=bl; i++)
   {
      a[i] = b[i];
   }
   printf("Copied String = %s\n\n", a);
}
//Reverse string
void reverse(char a[])
{
```

```
   int al = length(a);
   char temp;
   for(i=0, j=al-1; i<j; i++, j--)
   {
      temp = a[i];
      a[i] = a[j];
      a[j] = temp;
   }
   printf("Reverse String = %s\n\n", a);
}
```

OUTPUT:

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 1
Enter String: HELLO
Length of the String is 5
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 2
Enter String: Hello
Enter second String: good
Copied String = good

ENTER CHOICE (7 for Display Menu):
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 3
Enter String: Hello
Reverse String = olleH

ENTER CHOICE (7 for Display Menu):
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 4
Enter String: MADAM
String is Palindrome

ENTER CHOICE (7 for Display Menu):
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 5
Enter String: Hello
Enter substring String: Hel
The String B is substring of A and it occur 1 times

ENTER CHOICE (7 for Display Menu):
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 6
Enter String: Hello
Enter second String: Hello
Both Strings are Equal

ENTER CHOICE (7 for Display Menu):
```

Output

```
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 7
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu):
```

Output

```
<<-----------Menu------------>>
1. Length
2. Copy String into another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with Another
7. Display Menu
8. Exit
ENTER CHOICE (7 for Display Menu): 8


=== Code Execution Successful ===
```